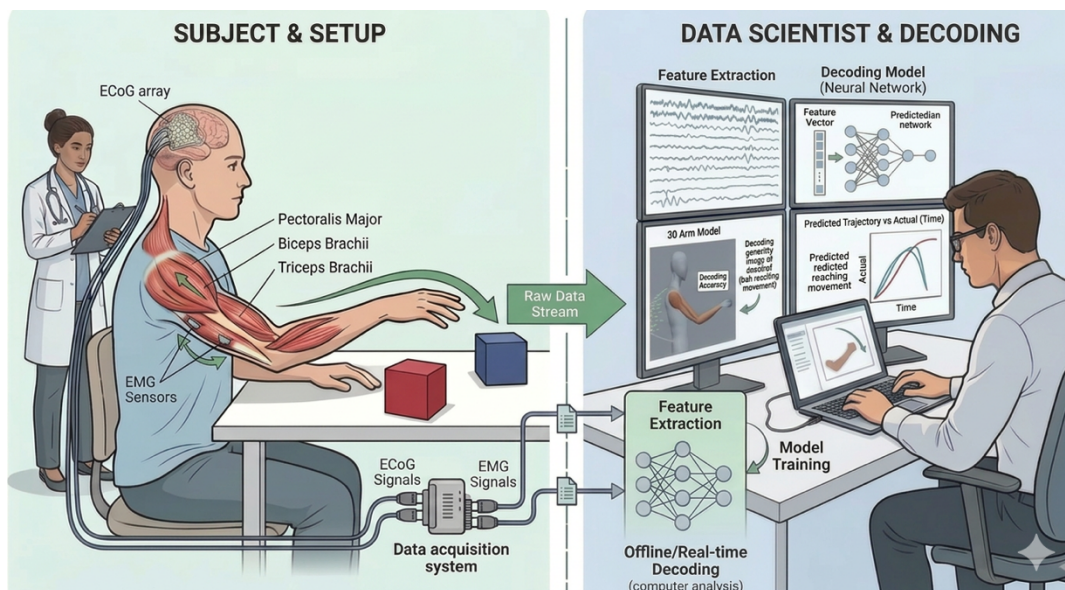


Machine Learning for Sensorimotor Neuroscience

Graduate Course Syllabus — 15 Weeks

A Unified Motor Control Thread: From Single-Joint Simulation to Neural Decoding



This course uses an artificially generated labeled reaching movements and cortical activity from either healthy participants or patients with a neurological injury. You'll develop data analysis skills by determining two things: whether the data reflects a healthy or injured participant, and which object (direction) was the target of each reaching movement.

Course Information

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Course Website

https://github.com/tarkeshsingh/ml-sensorimotor_neuroscience

Course Description

This graduate-level course introduces machine learning methods commonly used in neuroscience research, with all implementations in Python. Students build foundational programming skills (Weeks 1–3) then construct a multi-joint reaching dataset (Week 4) that serves as the unified testbed for every ML method in the course. By applying 10 different algorithms to the same progressively enriched dataset, students develop deep intuition for when and why each method succeeds or fails on real motor control data.

A distinctive feature of this course is the integration of modern AI tools into the scientific workflow. Students learn to use AI assistants to accelerate data analysis, debug code, generate visualisations, and iterate on ML pipelines — skills that are rapidly becoming essential in computational neuroscience research.

Prerequisites: Basic statistics, introductory programming experience recommended

Software: Python 3.x, Google Colab or Jupyter Notebook, scikit-learn, PyTorch

Learning Objectives

By the end of this course, students will be able to:

1. Implement and compare supervised and unsupervised machine learning methods (logistic regression, Naive Bayes, KNN, SVM, LDA, random forests, k-means, GMM, and neural networks) using Python and scikit-learn.
2. Design and evaluate ML pipelines for neuroscience data, including feature extraction, cross-validation (with emphasis on leave-one-subject-out), hyperparameter tuning, and performance metrics appropriate for clinical and BCI applications.
3. Critically assess which ML method is appropriate for a given dataset based on data size, feature type, interpretability requirements, and computational constraints — using the decision framework developed across the course.
4. Apply dimensionality reduction (PCA, LDA) and feature engineering (muscle synergies, temporal neural features) to transform raw neurophysiological signals into informative representations for classification.
5. Use AI tools (e.g., Claude, ChatGPT, Copilot) to accelerate ML analyses on experimental sensorimotor control data — including code generation, debugging, data visualisation, and iterative pipeline development — while maintaining scientific rigour and understanding of the underlying methods.
6. Train, diagnose, and interpret neural networks on neurophysiological data, including reading loss curves, applying regularisation, computing saliency maps, and understanding when deep learning adds value over classical methods.
7. Identify and articulate ethical considerations in clinical ML, including data privacy, algorithmic bias, and responsible reporting of results in neuroscience publications.
8. Communicate ML results clearly through visualisations, comparison tables, confusion matrices, and written interpretation grounded in motor control neuroscience.

Course Outline

Part I: Python Foundations & Motor Control Models (Weeks 1–3)

Week 1: Python Essentials, NumPy & Matplotlib

Python environment, NumPy arrays, and Matplotlib for scientific computing. Neuroscience application: simulate a single-joint reaching movement with triphasic EMG patterns, Euler integration, and bell-shaped velocity profiles.

Lab: EMG-driven point-to-point reaching simulation with parameter exploration.

Week 2: Data Analysis with Pandas & Seaborn

DataFrames, groupby aggregation, and statistical visualisation. Neuroscience application: multi-trial EMG and kinematic analysis across subjects and conditions, including healthy vs. impaired comparison.

Lab: Experimental data analysis with Pandas — violin plots, confidence intervals, and inter-subject variability.

Week 3: Object-Oriented Programming

Classes, inheritance, polymorphism, and abstract base classes for reusable analysis pipelines. Neuroscience application: three closed-loop motor control models (Equilibrium Point, Optimal Feedback Control, VITE) compared via a shared class hierarchy.

Lab: Motor controller class hierarchy with perturbation analysis and polymorphic comparison.

Part II: Building the Course Dataset (Week 4)

Week 4: Multi-Joint Reaching, Muscle Synergies & PCA

2-DOF shoulder–elbow model, 6-muscle EMG with synergy-based activation, and PCA for dimensionality reduction. The course dataset is constructed: 8 directions × 3 speeds × 20 subjects (10 healthy, 10 impaired) = 480 trials. Healthy subjects use 3 synergies; impaired use 2 (module merging). Varimax rotation for physiological interpretability.

Lab: Forward kinematics, EMG generation, PCA with scree plots, Varimax rotation, and the sklearn Pipeline API.

Part III: Model Selection & Workflow (Week 5)

Week 5: Logistic Regression & Model Evaluation

From unsupervised (PCA) to supervised learning. Logistic regression, regularisation (L1/L2), cross-validation (k-fold vs. LOSO), confusion matrices, ROC/AUC, and the bias–variance tradeoff. Establishes the baseline comparison table that every subsequent week extends.

Lab: Classification pipeline, LOSO vs. k-fold comparison, hyperparameter grid search, and learning curves.

Part IV: Supervised Learning Methods (Weeks 6–11)

Week 6: Generalized Linear Models — From Muscles to Neurons

GLM framework unifying logistic regression (classification) and Poisson regression (spike counts). Cosine tuning, population vectors, encoding vs. decoding. Adds ~80 simulated M1 neurons to the dataset; the comparison table now tracks EMG and neural features side by side.

Lab: Poisson GLM tuning curves, population vector decoding, neural vs. EMG feature comparison.

Week 7: Naive Bayes

Bayes' theorem, Gaussian Naive Bayes, the conditional independence assumption and why it fails on correlated muscles. PCA decorrelation rescues accuracy (65% → 85%). Generative vs. discriminative classifiers.

Lab: Base rate fallacy, class-conditional Gaussians, correlation-driven failure, and PCA rescue.

Week 8: K-Nearest Neighbors

Distance metrics (Euclidean, Mahalanobis), the curse of dimensionality, choosing k . KNN achieves 100% on EMG binary but collapses in high dimensions. Non-parametric vs. parametric tradeoffs.

Lab: KNN on raw vs. PCA-reduced features, distance metric comparison, and bias–variance as a function of k .

Week 9: Support Vector Machines

Maximum margin classification, soft margins (C parameter), the kernel trick (linear, polynomial, RBF). Margin as a robustness buffer against electrode drift. BCI pipeline framing: computational cost, prediction speed, confidence-gated decoding.

Lab: Linear and RBF SVM on neural data, kernel comparison, confusion matrices, and drift robustness analysis.

Week 10: Linear Discriminant Analysis

LDA as simultaneous classification and dimensionality reduction. Fisher’s criterion, between- vs. within-class scatter. LDA vs. PCA: supervised vs. unsupervised projection. Neural population trajectories in LDA space.

Lab: LDA vs. PCA projections, silhouette scores, neural trajectory visualisation, and transfer learning preview.

Week 11: Random Forests & Ensemble Methods

Decision trees, bagging, boosting, and random forests. Feature importance as a clinical biomarker discovery tool. RF achieves 100% on EMG binary with built-in interpretability. Comparison with SVM’s opaque boundaries.

Lab: RF classification, feature importance ranking, comparison with PCA loadings and SVM weights.

Part V: Unsupervised Learning (Week 12)

Week 12: K-Means Clustering & Gaussian Mixture Models

K-means algorithm, k-means++ initialisation, choosing k (elbow, silhouette — both fail on this data). GMM as soft clustering with per-cluster covariance. Clustering recovers 8 directions (ARI=0.92) but completely misses healthy vs. impaired. The ‘subtypes’ illusion: clusters in impaired data are directions, not patient subtypes.

Lab: K-means and GMM on neural and EMG data, ARI evaluation, covariance type comparison, and the variance hierarchy.

Part VI: From Neural Networks to Course Integration (Weeks 13–15)

Week 13: Artificial Neural Networks — Concepts

From logistic regression to MLPs: the single unit, hidden layers as learned feature detectors, relu activation, backpropagation as error correction. MLP matches but does not beat classical methods on 480 trials (~94% vs. LDA 95.4%). Hidden representations separate directions more cleanly (silhouette 0.621) than PCA or LDA. Regularisation matters less than data size on small datasets.

Lab: Build a perceptron, architecture sweep, training curves, and hidden representation visualisation.

Week 14: Temporal Features — The Features Matter More Than the Classifier

Temporal neural features (80 neurons × 50 time bins = 4,000 features) transform impairment detection from 70–89% to 99%. The central lesson: LR on temporal features (99.0%) beats MLP on mean rates (79.8%) by 19 points — richer features with a simpler model outperform fancier models on poor features. Practical diagnostics: reading loss curves to identify training failures, data augmentation that respects the signal (noise helps, jitter destroys the onset-delay signal), saliency maps for neuron–time interpretability, and drift robustness analysis. The decision framework flowchart: how to choose a method based on labels, feature type, data size, and interpretability needs.

Lab: Temporal feature comparison, training vs. validation curves, augmentation experiments, saliency maps, and drift analysis.

Week 15: Course Integration & Final Projects

The complete 10-method comparison table across all tasks, with discussion of each method's strengths and weaknesses. Decision framework flowchart: labels? feature type? data size? interpretability? Ethical considerations: data privacy, algorithmic bias, and responsible reporting. Handout: 'Beyond This Course' covering RNNs/LSTMs, CNNs, autoencoders, and transformers with PyTorch code.

Final project presentations: end-to-end ML pipeline on a neuroscience dataset of the student's choice.

Assessment

- Weekly lab notebooks (40%) — Jupyter notebooks with code, visualisations, and written interpretation.
- Midterm project (20%) — Comparison of 4+ methods on the course dataset with written analysis.
- Final project (30%) — End-to-end ML pipeline applied to a neuroscience dataset of the student's choice.
- Participation (10%) — In-class discussion, peer code review, and engagement with ethical case studies.

Weekly Readings

Each week includes one primary reading from the motor control, BCI, or computational neuroscience literature that connects the ML method to its neuroscience application. Students should read the paper before the lecture.

Required readings before first day of class:

Halilaj, E., Rajagopal, A., Fiterau, M., Hicks, J. L., Hastie, T. J., & Delp, S. L. (2018). Machine learning in human movement biomechanics: Best practices, common pitfalls, and new opportunities. *Journal of Biomechanics*, 81, 1-11.

- **Week 1: Python & Biomechanics**

Flash T, Hogan N. The coordination of arm movements: an experimentally confirmed mathematical model. *Journal of Neuroscience*, 5(7), 1688–1703, 1985.

The minimum-jerk model that produces the bell-shaped velocity profiles simulated in the lab.

- **Week 2: Experimental Data Analysis**

Sainburg RL, Ghilardi MF, Poizner H, Ghez C. Control of limb dynamics in normal subjects and patients without proprioception. *Journal of Neurophysiology*, 73(2), 820–835, 1995.

Multi-trial kinematic analysis of reaching — the kind of data students analyse with Pandas.

- **Week 3: Motor Control Models**

Todorov E, Jordan MI. Optimal feedback control as a theory of motor coordination. *Nature Neuroscience*, 5(11), 1226–1235, 2002.

The OFC model implemented in the lab's class hierarchy.

- **Week 4: Muscle Synergies & PCA**

Tresch MC, Cheung VCK, d'Avella A. Matrix factorization algorithms for the identification of muscle synergies: evaluation on simulated and experimental data sets. *Journal of Neurophysiology*, 95(4), 2199–2212, 2006.

Directly motivates the PCA and NMF analyses on the course dataset.

- **Week 5: Classification & Cross-Validation**

Glaser JJ, Benjamin AS, Farhoodi R, Kording KP. The roles of supervised machine learning in systems neuroscience. *Progress in Neurobiology*, 175, 126–137, 2019.

Comprehensive review of ML in neuroscience — covers classification, cross-validation pitfalls, and when to use which method.

- **Week 6: Neural Encoding & Decoding**

Georgopoulos AP, Kalaska JF, Caminiti R, Massey JT. On the relations between the direction of two-dimensional arm movements and cell discharge in primate motor cortex. *Journal of Neuroscience*, 2(11), 1527–1537, 1982.

The foundational cosine tuning paper that the course dataset is built on.

- **Week 7: Bayesian Decoding**

Zhang K, Ginzburg I, McNaughton BL, Sejnowski TJ. Interpreting neuronal population activity by reconstruction: unified framework with application to hippocampal place cells. *Journal of Neurophysiology*, 79(2), 1017–1044, 1998.

Bayesian decoding of neural population activity — the neuroscience counterpart to Naive Bayes.

- **Week 8: Dimensionality & Distance**

Cunningham JP, Yu BM. Dimensionality reduction for large-scale neural recordings. *Nature Neuroscience*, 17(11), 1500–1509, 2014.

Why dimensionality reduction matters for neural data — connects KNN's curse of dimensionality to neural manifolds.

- **Week 9: SVMs & BCI**

Quitadamo, L., Cavrini, F., Sbernini, L., Riillo, F., Bianchi, L., Seri, S., & Saggio, G. (2017). Support vector machines to detect physiological patterns for EEG and EMG-based human–computer interaction: a review. *Journal of Neural Engineering*, 14(1), 011001.

The BCI review that contextualises SVM's role in motor intention decoding.

- **Week 10: LDA & Population Dynamics**

Eluchans M, Maselli A, Lancia GL, Pezzulo G. Eye and hand coarticulation during problem solving reveals hierarchically organized planning. *Journal of Neurophysiology*, 134(3), 985–997, 2025.

Modern application of dimensionality reduction to reveal hierarchical motor planning from eye–hand coordination data.

- **Week 11: Ensemble Methods & Biomarkers**

Saeb S, Lonini L, Jayaraman A, Mohr DC, Kording KP. The need to approximate the use-case in clinical machine learning. *GigaScience*, 6(5), gix019, 2017.

Why evaluation methodology matters more than model complexity in clinical ML.

- **Week 12: Clustering & Unsupervised Discovery**

Gallego JA, Perich MG, Miller LE, Solla SA. Neural manifolds for the control of movement. *Neuron*, 94(5), 978–984, 2017.

Low-dimensional neural manifolds — the structure that clustering and PCA discover in population activity.

- **Week 13: Neural Networks**

Moon, S., Ahmadnezhad, P., Song, H.-J., Thompson, J., Kipp, K., Akinwuntan, A. E., & Devos, H. (2020). Artificial neural networks in neurorehabilitation: A scoping review. *NeuroRehabilitation*, 46(3), 259–269.

Demonstrates use of ANNs for clinical decision-making of complex datasets in neurorehabilitation.

- **Week 14: Temporal Features & Feature Engineering**

Pandarinath C, O'Shea DJ, Collins J, et al. Inferring single-trial neural population dynamics using sequential auto-encoders. *Nature Methods*, 15, 805–815, 2018.

LFADS — temporal neural data analysis that motivates preserving the full time-series rather than collapsing to mean rates.

- **Week 15: Integration & Ethics**

Obermeyer Z, Powers B, Vogeli C, Mullainathan S. Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464), 447–453, 2019.

A landmark study of algorithmic bias in healthcare ML — required reading for the ethics discussion.

Reading for end of semester:

Myszczyńska, M. A., Ojamies, P. N., Lacoste, A. M. B., Neil, D., Saffari, A., Mead, R., Hautbergue, G. M., Holbrook, J. D., & Ferraiuolo, L. (2020). Applications of machine learning to diagnosis and treatment of neurodegenerative diseases. *Nature Reviews Neurology*, 16(8), 440–456.

Recommended Texts

- Hastie, Tibshirani & Friedman — The Elements of Statistical Learning (Springer, 2009)
- Géron — Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (O'Reilly, 3rd ed.)
- Dayan & Abbott — Theoretical Neuroscience (MIT Press, 2001)
- scikit-learn documentation: <https://scikit-learn.org>
- Neuromatch Academy computational neuroscience tutorials: <https://compneuro.neuromatch.io>

Note on AI-Assisted Content Development. *The lecture notes, lab notebooks, and supporting materials for this course were developed with the assistance of an AI language model (Claude Opus 4.6, Anthropic). All content has been reviewed and checked for accuracy, but AI-generated materials may still contain errors, inconsistencies, or inaccurate statements. Students are encouraged to think critically about the material and consult the primary references listed in the weekly readings. If you identify any errors, please report them to Tarkesh Singh at tsingh@psu.edu so they can be corrected for future use.*

Syllabus subject to revision. Adjustments may be made based on class progress and interests.